Rotaract Club of Xiamen University Malaysia

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	PostDepresSQL
Project / Solution Name	SkillShift AI
Name (Leader)	SARVEZ ELANGOVAN
Email & Phone (Leader)	sarvezelangovan69@gmail.com - +60167229286
University Affiliation(s)	Tunku Abdul Rahman University of Management and Technology (TARUMT)
Live Prototype URL	https://skillshift-ai.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/TheRealSRZ/skillshift-ai

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

SkillShift AI uses a modular web architecture deployed through Replit.

The React and TypeScript frontend collects fictional learner profiles and communicates with Python FastAPI endpoints. Backend services validate inputs, load curated career/skill catalogs, and calculate weighted career-readiness scores, skill gaps, and hypothetical scenario comparisons. The What-If Simulator previews the effect of learning additional skills without changing the original profile.

A separate evidence layer presents attributed workforce observations and source dates where available. Learning-roadmap generation selects courses from a curated, approved catalog and returns a structured plan with relevant learning activities and estimated duration.

Supabase provides authentication and PostgreSQL storage for user-owned account information, profiles, and saved results. Server-side verification protects authenticated backend operations.

Optional Qwen inference through RunPod personalizes the selection or ordering of approved learning resources. Its output is validated against the course catalog; if inference is unavailable, the deterministic roadmap remains accessible.

Replit hosts the compiled frontend and FastAPI service under one public HTTPS address. Private service credentials remain in server-side environment settings rather than frontend source code.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Backend programming language	PSF License
React 19	Component-based web UI	MIT
TypeScript — installed version	Frontend type checking	Apache-2.0
Vite — installed version	React build tool	MIT
FastAPI — installed version	REST API backend	MIT
Pydantic 2.x	Input validation and schemas	MIT
Uvicorn — installed version	ASGI application server	BSD-3-Clause
Supabase JavaScript SDK — installed version	Authentication and database client	MIT
Recharts — installed version	Data charts	MIT
Tailwind CSS — only if installed	Styling	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Supabase Authentication API	User registration, login, token verification and account management	Actual Supabase plan — confirm
Supabase Data API	Save and retrieve user-owned profiles, analyses and roadmaps	Actual Supabase plan — confirm
RunPod Serverless API	Optional Qwen-based roadmap personalization	Sponsor-provided credits, if used
Google OAuth via Supabase	Google sign-in	Only if actually configured and used

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Host the React production build and FastAPI backend	Actual deployment type and region
Supabase	Authentication and managed PostgreSQL database	Actual plan and project region
RunPod	Optional GPU inference endpoint	Actual Serverless plan, GPU and region
GitHub	Source-code hosting and version control	Free/paid plan as actually used

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen — exact checkpoint to be inserted	Qwen model publisher; hosted on RunPod	Pretrained model used for optional approved-course personalization	Verify the exact model's license

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Curated career and skills catalogs	Team-maintained data/ files; underlying sources to be cited	Curated / partly synthetic	Team content; original-source terms apply	No
Learning-resource catalog	Team-maintained course list, linking to providers such as Microsoft Learn	Curated public metadata	Provider terms apply	No
DOSM occupation-level online job-offering observations	Exact DOSM publications and URLs from Member 4	Public	Verify DOSM reuse/attribution terms	No
AI-adoption scenario multipliers	Team-developed scenario configuration	Synthetic	Team-developed	No
Fictional user/demo profiles	Team-developed examples	Synthetic	Team-developed	No

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Other O*NET/OECD/TalentCorp evidence	Include separately only when actually used	Public/reference data	Source-specific terms	No, if only aggregate occupational data

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI ChatGPT (GPT models)	Technical planning, code assistance, debugging, regression-test development, security review and documentation drafting	Major
Replit Agent	Used to deploy our project to a public, accessible URL.	Minor
Anthropic's Claude	Frontend design and development.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI development tools supported brainstorming, implementation assistance, debugging, documentation and test preparation.

The team reviewed suggested changes against its architecture, functional requirements and approved data sources, adapting code before integration. Backend functionality was checked through automated tests, with the latest reported local run passing 165 tests.

The team also reviewed input validation, allowed course identifiers, authentication boundaries and failure handling. We recognize that automated tests do not establish complete security or real-world forecasting accuracy. Registered members retain responsibility for technical decisions, originality, licensing, and the accuracy of all submitted materials.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Replit Core / application-hosting platform	Published SkillShift AI frontend and FastAPI backend
RunPod	GPU credits / Serverless compute	Include if Member 3 used sponsor credits or RunPod compute for actual inference

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the deployed hosting environment and is essential to the prototype's current public accessibility, although the application architecture could technically be deployed through another compatible web-hosting provider.

RunPod supports optional GPU-based learning-roadmap personalization. It is not essential to the platform's deterministic career-matching, scenario-analysis, or approved-course roadmap features. If GPU inference is unavailable, users can still receive explainable career matches and relevant learning recommendations without calling the language model.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

SkillShift AI processes learner-entered skills, educational information, interests and target career preferences. Public demonstrations use fictional learner profiles rather than real CVs.

When a user registers or signs in, Supabase handles authentication information, including email addresses, and stores account-associated data. Users may save profiles, analyses and learning roadmaps. Access to user-owned database records is restricted through authentication and Row Level Security policies.

Private backend credentials are stored in deployment environment settings. API inputs are validated, and authenticated operations verify user identity.

When optional RunPod personalization is enabled, relevant roadmap context and approved course candidates may be transmitted to the inference endpoint. The system should avoid including unnecessary personal identifiers in inference requests.

Saved information remains subject to application deletion controls and the relevant cloud providers' retention policies; no universal fixed retention period is claimed. The prototype is not a substitute for a production privacy audit.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The platform uses a limited curated career and course catalog; some skills or occupations may not have sufficient approved learning resources or comparable Malaysian labor-market evidence. Scenario multipliers are hypothetical modelling assumptions, not validated employment forecasts. Career-readiness scores measure weighted skill overlap rather than hiring probability.

Optional language-model inference may produce inaccurate, incomplete or inconsistent outputs. The backend restricts recommended course identifiers to the approved catalog and retains a deterministic fallback when inference is unavailable or invalid.

RunPod may experience cold starts, execution delays, timeouts or exhausted credits. Public AI access can remain disabled until operational quota persistence is verified.

Account and API protections include input validation, authenticated access controls and request limits; however, a hackathon MVP has not undergone a complete independent production security assessment.

Future improvements include broader evidence coverage, validated forecasting methods, accessibility evaluation and durable distributed quota controls.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[24th October 2026]